

Face Verification System Card

Group Name: ML Verifier

Members: Moataz Osama Saadeldin Abdelaziz, Ankan Roy

Overview

This is a face verification system that determines whether two input images belong to the same individual. It operates by taking two images and computing a similarity score between them. Based on a predefined threshold, the system outputs both a binary decision and a confidence score indicating how strongly the decision is supported. The system was developed using the Labeled Faces in the Wild (LFW) dataset and evolved through multiple milestones, culminating in an embedding based verification pipeline with a command line interface, Docker support, and concurrent load testing.

Intended Use

This system is intended for face verification tasks where the goal is to determine whether two images contain the same person. Scenarios include, but are not limited to identity matching or duplicate detection. The system is not designed for face identification across large databases, real time surveillance or high stakes security applications. It should not be used in environments where incorrect decisions could lead to significant harm, such as law enforcement or biometric authentication without additional safeguards.

System Description

The system follows a structured inference pipeline. It first preprocesses the input images by loading and converting them into RGB format. It then generates embeddings using the InsightFace buffalo_s model, which produces a numerical representation of each face. The system computes cosine similarity between the two embeddings and compares the result to a fixed threshold to determine whether the images represent the same individual. In addition to the binary decision, the system computes a confidence score based on how far the similarity score lies from the decision threshold. It also records latency for each inference request.

Operating Threshold

The system uses a fixed cosine similarity threshold of 0.29 to make verification decisions. If the similarity score exceeds this threshold, the system predicts that the images belong to the same person. Otherwise, it predicts that they belong to different individuals. This threshold was selected based on a sampled validation sweep, replacing an earlier baseline threshold of 0.35. The choice reflects a trade off between false acceptances and false rejections observed during validation.

Key Metrics

The system was evaluated through validation sweeps and benchmarking pipelines, focusing on threshold calibration and performance consistency. The system emphasizes reproducible evaluation and threshold optimization. It also tracks inference latency and supports concurrent load testing, which provides insight into performance under multiple simultaneous requests. Confidence scores are reported for each prediction, although these scores are not probabilistic and should not be interpreted as calibrated likelihoods. For the final embedding based system using the operating threshold of 0.29, the reported metrics are:

Accuracy: 0.99944

Precision: 1.00000

Recall / TPR: 0.99774

Specificity: 1.00000

F1 score: 0.99887

False positive rate / FPR: 0.00000

Balanced accuracy: 0.99887

True positives: 2209

True negatives: 6642

False positives: 0

False negatives: 5

Confidence Interpretation

The system produces a confidence value between 0 and 1 that reflects how far the similarity score is from the decision boundary. This value is computed using a margin based function that scales the absolute difference between the score and the threshold. A confidence value near zero indicates that the similarity score is very close to the threshold, meaning the decision is uncertain. A value near one indicates that the score is far from the threshold, meaning the decision is more clearly supported. It is important to note that this confidence score is not a true probability and should not be interpreted as such.

Limitations

The system relies on a lightweight embedding model, which prioritizes speed over maximum accuracy. As a result, it may struggle with difficult cases such as low resolution images, occlusions, extreme lighting conditions, or unusual facial angles.

Because the system uses a fixed threshold, it does not adapt dynamically to different data distributions or deployment contexts. Performance may degrade when applied to data that differs significantly from the training and validation dataset.

Additionally, the system does not include explicit face detection robustness checks or alignment guarantees beyond what is provided by the embedding model. This can lead to inconsistent embeddings when faces are poorly framed or partially visible.

The system also depends on successful face detection before embedding extraction. If face detection fails, the system may return a non-match or an unreliable result rather than a meaningful face comparison. When multiple faces appear in an image, the system selects the largest detected face, but this may not always be the intended person.

Failure Modes

Failure modes refer to the specific ways in which the system can produce incorrect or unreliable outputs. In a face verification system, this typically includes false positives, where two different individuals are incorrectly classified as the same person, and false negatives, where two images of the same person are incorrectly classified as different. Other failure modes may arise when the similarity score is close to the threshold, leading to unstable decisions, or when the input images contain noise, blur, or non-face content that disrupts embedding generation. Understanding failure modes is important because it helps identify conditions under which the system should not be trusted or should defer to human review.

Fairness Related Risks

Fairness related risks refer to the possibility that the system performs differently across demographic groups, such as variations in accuracy based on age, gender, or ethnicity. These disparities can arise from imbalances or biases in the training dataset or from limitations in the embedding model. Although this project does not explicitly evaluate fairness, such risks are important to consider in any face verification system. Uneven performance across groups can lead to systematic disadvantages or harms in real world applications.

Deployment Constraints

The system is designed to be deployed as a command line interface application and can be containerized using Docker for portability. It requires access to the embedding model and sufficient computational resources to process image inputs and generate embeddings efficiently. The system supports concurrent inference using multithreading, but performance will depend on the available hardware and the number of simultaneous requests. It is best suited for batch or low to moderate throughput scenarios rather than large scale, real time deployments. Because the system uses a fixed threshold and

margin based confidence, deployment environments should carefully validate performance on their specific data before relying on the system for decision-making. The final system was profiled on CPU for the required baseline. The profiling report is located at reports/profiling_report.md, and the profiling CSV is located at reports/profiling_cpu_summary.csv.

Reproducibility

The final release is reproducible from a clean clone using the README and reproducibility checklist.

Important files:

README.md

reports/system_card.md

reports/profiling_report.md

reports/reproducibility_checklist.md

src/config.py

Final Git tag:

v1.0-final